Study #1: **Harnessing Synergies: Integrating AI Literacy and Data Analytics into Cybersecurity Training Programs for High School Students**

What were the main purposes of the study?
** In today's educational landscape, there is growing concern that education can sometimes be too generalized. Moreover, as Willingham (2009) mentioned, many specializations students pursue in higher education may not necessarily require a broad foundation in subjects such as history, language arts, or music; this includes many STEM subjects such as most engineering and healthcare majors.

Nevertheless, things cannot be said the same for cybersecurity issues in this fast-paced internet society. With issues such as cyberbullying, online fraud, pornography, racial discrimination, and gambling on the rise, one of the vital measures to be taken is to improve digital preparedness among internet users (Rahman et al., 2020). Therefore, a holistic educational approach is necessary, incorporating cybersecurity awareness and ethical considerations into the curriculum. By integrating AI and data in cybersecurity education, we can ensure that students not only understand the technologies they interact with but also the ethical implications and risks associated with them.

What are the research questions?
**Here are some questions that can be used:

- How does integrating AI literacy improve students' understanding of cybersecurity concepts?
  *Answers can be in terms of literacy review or by conducting a paid study on high school students by doing experiments in the form of a game
- How can the integration of AI and cybersecurity be implemented inside the K-12 education curricula?
  *Cross-disciplinary learning, AI and Cybersecurity Projects, Capture the Flag (CTF) Challenges
- What role does data analytics play in enhancing cybersecurity problem-solving skills in high school students?
  *The answer can be derived as a part of the conclusion of the question above
- How effective is the combined approach of AI and data analytics in preparing students for future cybersecurity challenges?
  *This is the final result of the whole research ;)
- What are the key barriers and opportunities in implementing AI literacy and data analytics into high school cybersecurity programs?
  *The answer can be simple; such as the high logical skill required, lack of resources, inadequacies of the lecturers/ teachers, and many more

Who was studied (sample, sample size, special characteristics)?
**
Sample: High school students, aged 15-18, from various schools in a well-developed country (Singapore) where the students have a better understanding and more resources, and a growing country (Indonesia) where the students have less knowledge and fewer resources

Sample Size: Approximately 100-150 students from both well-developed and growing schools to ensure a diverse set of perspectives with a ratio of 1:2 to create a bias where developed countries have half the impact of the developing countries.

Special Characteristics: Students with basic exposure to computer science or cybersecurity-related subjects, though no prior knowledge of AI or data analytics is required.
The sample may also include cybersecurity and AI teachers as secondary participants to understand their perspectives on the integration.

How were participants selected?
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Purposive Sampling: Schools that offer cybersecurity or technology programs are more targeted.

Voluntary Participation: Students who expressed interest in technology or cybersecurity were encouraged to participate.
Demographic Diversity: Efforts were made to include students from both developed and developing countries, ensuring a wide range of socioeconomic backgrounds.

Collaboration with Schools: Partnerships with schools with no cybersecurity or technology programs were also established to ensure a broad and representative sample.

To what conditions, if any, were participants exposed?
**The participants are exposed to a few of these modules and exercises:

- **AI Literacy Modules**: Participants were exposed to modules on AI fundamentals, covering topics such as machine learning, AI ethics, and real-world applications from a few hours of remote training as well as some data analytics exercises where students worked on projects using data analytics tools to identify patterns in cybersecurity threats.
- **Hands-on Cybersecurity Challenges**: Students applied AI and data analytics skills in simulated cybersecurity scenarios, testing their ability to detect and prevent digital threats through offline meetings.
- **Control Group**: A separate group of students followed traditional cybersecurity training without AI and data analytics integration to assess the differences in learning outcomes.

What type of research design was used?
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Mixed Methods Design:

By combining both quantitative (numerical data) and qualitative (non-numerical data) approaches through surveys, tests, or other forms of data collection to provide a more comprehensive understanding of the research problem. Quantitative data can offer **measurable evidence** of the program's effectiveness, while qualitative data can provide **insights** into students' **thoughts and feelings** about the curriculum.

Quasi-Experimental Design:

By conducting pre- and post-tests to assess students' knowledge and skills before and after the curriculum is introduced to see if there's a measurable impact regarding the understanding of AI, data analytics, and cybersecurity improved as a result of the new curriculum.

Intervention-Based Design:

Explanation: Here, the introduction of AI and data analytics into the curriculum is treated as an **intervention**. One group of students (the experimental group) receives this updated curriculum, while a control group continues with traditional cybersecurity training.

Purpose: This allows for a comparison between students who received the intervention and those who did not, helping to isolate the effect of the new curriculum.

What type of research analysis was used?
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- **Quantitative Analysis**: Statistical analysis of pre- and post-test results, using tools like SPSS to measure improvements in student performance or time recordings to show the improvements in terms of efficiency as well.
- **Qualitative Analysis**: Thematic analysis of student interviews and feedback to assess their engagement and understanding of AI and cybersecurity concepts.
- **Comparative Analysis**: Comparing the learning outcomes of the intervention group (AI + data analytics) with the control group (traditional cybersecurity training).
- **Descriptive Statistics**: Summarize all of the other analyses above to create data for highlighting trends, mean scores, and standard deviations in performance between groups.

What are the expected outcomes and conclusions?
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Improved Cybersecurity Skills: Students exposed to AI and data analytics are expected to show a deeper understanding of cybersecurity threats and stronger problem-solving skills.

Increased Awareness of AI Ethics: The integrated curriculum is likely to raise awareness of ethical concerns in AI and data use, particularly in cybersecurity contexts.

Enhanced Critical Thinking: Students are anticipated to develop better critical thinking abilities, especially in detecting and preventing digital threats using data-driven approaches.

Long-Term Curriculum Adoption: The study may conclude that integrating AI and data analytics into cybersecurity training should be adopted on a wider scale in high school curricula.
Challenges in Implementation: Potential barriers such as resource limitations, teacher readiness, and curriculum adaptation may be highlighted as areas for future focus.

Willingham, D. T. (2009). Why don't students like school? In *AMERICAN EDUCATOR* (pp. 4–5).https://www.nottinghamschoolstrust.org.uk/wp-content/uploads/2019/06/Why-students-dont-like-school-.pdf
Rahman, N. a. A., Sairi, I. H., Zizi, N. a. M., & Khalid, F. (2020). *The importance of cybersecurity education in school*.
https://www.semanticscholar.org/paper/The-Importance-of-Cybersecurity-Education-in-School-Rahman-Sairi/86ffd5ed7c2dd7a53fa9797250b8270556faef3e#citing-papers

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Example

This initial study investigates the various sources of knowledge that may influence first-generation and continuing-generation college students in Singapore's engineering learning. College students' learning of engineering may be influenced by their families, communities, and work and school-related acquaintances sharing knowledge and skills. The purpose of this study is to determine how, and which sources of knowledge can assist a first-generation university student in adapting to the engineering course, seeking assistance from peers and those around them, and applying skills learned in the industrial aspect. This study employs a validated survey based on the ten latent constructs: tinkering knowledge from home and work, connecting experiences, networks from family members, college friends, colleagues, and neighborhood friends, perspective-taking, reading people, and mediating capability. The survey also included the constructs of engineering performance and competence beliefs. A quantitative research method was then used to analyze the data. The study's findings aim to gain a more comprehensive understanding of how engineering students learn, allowing educators to use this knowledge to improve teaching methodologies and materials to provide better learning experiences for engineering students.

Abstract points

Purpose: The study investigates the integration of AI literacy and data analytics into cybersecurity training programs for high school students in Singapore and Indonesia. It aims to assess how these integrations enhance students' understanding of cybersecurity concepts, problem-solving skills, and ethical considerations related to AI and data use.

Participants: The study involved approximately 100-150 high school students aged 15-18 from (Singapore). The sample included students with basic exposure to computer science or cybersecurity but no prior knowledge of AI or data analytics.

Selection Method: Purposive sampling was employed, targeting schools with cybersecurity or technology programs, and encouraging voluntary participation from students interested in these fields. Efforts were made to ensure a diverse demographic representation from both developed and developing countries.

Exposure: Participants were exposed to AI literacy modules, hands-on data analytics exercises, and simulated cybersecurity challenges. A control group received traditional cybersecurity training without AI and data analytics to compare outcomes.

Design: The study used a mixed-methods, quasi-experimental design with pre- and post-tests to measure the impact of the integrated curriculum. An intervention-based design was implemented, with one group receiving the updated curriculum while the control group followed traditional training.

Findings: The expected outcomes included improved problem-solving skills, a deeper understanding of cybersecurity threats, increased awareness of AI ethics, and enhanced critical thinking abilities. The study also identified challenges in implementation, such as resource limitations and teacher readiness, which may need to be addressed for long-term curriculum adoption

Harnessing Synergies: Integrating AI Literacy and Data Analytics into Cybersecurity Training Programs for High School Students

Abstract

With the growth of issues regarding cyberbullying, online fraud, pornography, racial discrimination, and gambling on the rise, one of the vital measures to be taken is to improve digital preparedness among internet users (Rahman et al., 2020). Lamentably, even if many parties are conducting webinars, conventional methods cannot secure the (Personally Identifiable Information) PII and (Sensitive Personally Identifiable Information) SPII of countless internet users. This is due to job burnout in the cybersecurity field driven by the rapid advancement of technology, the intense operational requirements, and the constant emergence of threats, all exacerbated by fragmented security solutions and a lack of integration among products (Platsis, 2019). Therefore, this study aims to investigate the synergies between the integration of AI literacy and data analytics into cybersecurity training programs for high school students in Singapore. Moreover, this study assesses how integrating AI and data analytics enhances students' understanding of cybersecurity concepts, problem-solving skills, and even ethical considerations. In terms of participants, the study engages the research with a diverse group of 50-75 high school students aged 15-18, drawn from various schools across Singapore, both well-equipped schools and those without specialized training in these fields. The ratio between male and female participants is also kept the same as 50:50. These selected students are also differentiated based on their cultural backgrounds and prior knowledge levels; some had exposure to AI, data analytics, or even cybersecurity, while others had none. By including participants, the study hopes to ensure a comprehensive, inclusive approach; giving an evaluation of how AI and data analytics can be integrated into cybersecurity education, regardless of the students' prior experiences.

This study mainly utilizes validated surveys based on 8 latent pointers, which are connections with existing knowledge, ethical awareness, engagement with the latest curriculum, technology readiness, perception towards cybersecurity, learning styles, challenges, and future opportunities. A qualitative analysis was then applied to the data to determine the pattern which then would be utilized for keyword analysis that emerged in students' understanding, engagement, and challenges. This approach is implemented in order to help identify gaps in curriculum design, technology readiness, and ethical considerations that need addressing to better integrate AI and data analytics into high school cybersecurity education. The research's findings plan to determine the level of understanding and expectancies of high school students regarding the current cybersecurity threats, data analytics competencies, ethical AI, and their potential to be implemented inside their curriculum. Moreover, this study also aims to determine the challenges and issues with the implementation of cybersecurity with AI and data analytics inside the current curricula, such as the inadequacies of the available resources and the lecturers which need to be addressed in the future. Overall, this whole study aims to improve and establish cybersecurity education in schools by integrating it with innovative technologies through AI and data analytics to develop high school students to face the future digital world.

Keywords: Personally Identifiable Information (PII), Sensitive Personally Identifiable Information (SPII), Artificial Intelligence (AI), Data Analytics, Cybersecurity, Problem-solving, ethical AI

Resources

Platsis, G. (2019). The Human Factor: Cyber Security's Greatest Challenge. In Cyber Law, Privacy, and Security: Concepts, Methodologies, Tools, and Applications (pp. 1-19). IGI Global.

Rahman, N. a. A., Sairi, I. H., Zizi, N. a. M., & Khalid, F. (2020). *The importance of cybersecurity education in school*. https://www.semanticscholar.org/paper/The-Importance-of-Cybersecurity-Education-in-School-Rahman-Sairi/86ffd5ed7c2dd7a53fa9797250b8270556faef3e#citing-papers

1. Connections with Existing Knowledge:

- Have you ever heard the word artificial intelligence (AI), cybersecurity, or data analytics before participating in this program, maybe through proper education or movies?
- How would you describe any of the previous experiences you've had with cybersecurity, AI, or data analytics, either in school or outside in 1 sentence?

2. Ethical Awareness:

- Do you think ethical considerations when working with AI technologies in cybersecurity are needed, if yes/no, why?
- What are some ethical concerns you foresee when integrating AI and data analytics into cybersecurity practices?

3. Engagement with the Latest Curriculum:

- How relevant do you find the content of the training program to the current issues in cybersecurity, such as cyberbullying or online fraud?
- How would you feel if the implementation of AI and data analytics were placed in your current curriculum?

4. Technology Readiness:

- How confident are you in using AI-based tools and data analytics technologies if you are currently hacked
- How well-equipped do you feel your school is in terms of providing the resources necessary for AI and cybersecurity education?

5. Perception Towards Cybersecurity:

- How important do you think cybersecurity skills are for your future career, especially in conjunction with AI and data analytics?
- In your opinion, is there a role that AI can play in order to strengthen the current cybersecurity measures?

6. Learning Styles:

- Which methods of learning (e.g., hands-on activities, lectures, group discussions) did you find most effective in understanding concepts?
- Did you find any particular approach (visual, auditory, practical) more helpful in understanding complex topics?

7. Challenges:

- Were there any topics or technologies discussed during the program that you think will be the hardest for you to learn?
- Do you have anything in mind when it comes to the resources needed to learn and master these skills in your school?

8. Future Opportunities:

- How interested are you in further pursuing education or a career in cybersecurity, particularly involving AI and data analytics?
- What areas of AI and cybersecurity would you like to explore more deeply in future programs?
- Do you feel the program has prepared you to deal with real-world cybersecurity challenges using AI and data analytics?

Submitted version

Title: Harnessing Synergies: Integrating Artificial Intelligence Literacy into Cybersecurity Training Programs for High School Students (Work-in-Progress)

Abstract: In this work-in-progress, the rise of cyberbullying, online fraud, racial discrimination, and gambling underscores the urgent need for digital awareness among Internet users (Rahman et al., 2020). Lamentably, even if many parties are conducting webinars, conventional methods cannot secure the Personally Identifiable Information (PII) and Sensitive Personally Identifiable Information (SPII) of countless Internet users. This is due to job burnout in the cybersecurity field, driven by technology's rapid advancement, intensive operational requirements, and the constant emergence of threats, exacerbated by fragmented security solutions and a lack of integration among products (Platsis, 2019). Therefore, this study aims to investigate the integration of artificial intelligence (AI) literacy into cybersecurity training programs for polytechnic students in Singapore. Moreover, this study assesses how AI integration enhances students' understanding of cybersecurity concepts, problem-solving skills, and even ethical considerations. Lastly, this study aims to determine the challenges and issues with the implementation of cybersecurity with AI within the current curricula. The following research questions guide the aim of the study: (a) What are the current levels of AI literacy and cybersecurity literacy among high school students? (b) What are the students' perceptions of the limitations of integrating AI literacy into cybersecurity training programs? This study engages a diverse group of 16 high school students aged 15-18 years old, selected from two schools in Singapore. The study aims to ensure a comprehensive and inclusive approach; giving an evaluation of how AI literacy could be integrated into cybersecurity education, regardless of the students' prior experiences. A qualitative research approach and analysis were applied to the semi-structured interview data to determine a pattern, which then would be utilized for keyword analysis that emerged in students' understanding, engagement, and challenges. This approach helps identify gaps within the curriculum design, technology readiness, and ethical considerations that need addressing, for better AI integration into high school cybersecurity education. The expected outcomes may contribute to determining the level of understanding and expectancies of high school students regarding current cybersecurity threats, ethical AI, and their potential to be implemented inside their curriculum. To conclude, while the study data collection is still in progress, this study may help improve and establish cybersecurity education in schools by integrating it with innovative technologies through AI and data analytics in order to better equip high school students for the digital world. Further elaboration of practical and theoretical implications, along with future recommendations, will be discussed.

Keywords: Artificial intelligence, polytechnic students, cyber-security, perceptions, K-12

For the ASEE submission, the study will be about only students' perceptions

For the manuscript & URECA project, the study will be about students' and teachers' perceptions (genders are equally splitted)

- (a) What are the current levels of AI literacy and cybersecurity literacy among high school students and teachers? (b) What are the students' and teachers' perceptions of the limitations of integrating AI literacy into cybersecurity training programs? This study engages a diverse group of 32 participants from NYP school (16 high school students aged 15-18 years old [8 females, 8 males] and 16 teachers [8 females, 8 males]) in Singapore.